Channel Estimation for Extremely Large-Scale Massive MIMO:
Far-Field, Near-Field, or Hybrid-Field?

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1 Introduction

Extremely large-scale massive multiple-input multiple-output (XL-MIMO) equips a base station (BS) with hundreds of antennas. The resulting aperture offers high array gain and spatial resolution for future 6G systems, but it also creates a difficult channel-estimation problem: the channel dimension grows with the antenna count, while sending one orthogonal pilot per antenna would consume an impractical amount of time and spectrum.

Wei and Dai identify a second problem that is specific to very large apertures [1]. Conventional massive-MIMO estimation assumes that all relevant scatterers are far from the array, so their waves are planar. In XL-MIMO, the Rayleigh distance can extend tens of metres; nearby scatterers then produce spherical waves. A practical channel can therefore contain far-field and near-field paths at the same time. The paper calls this the *hybrid-field* condition.

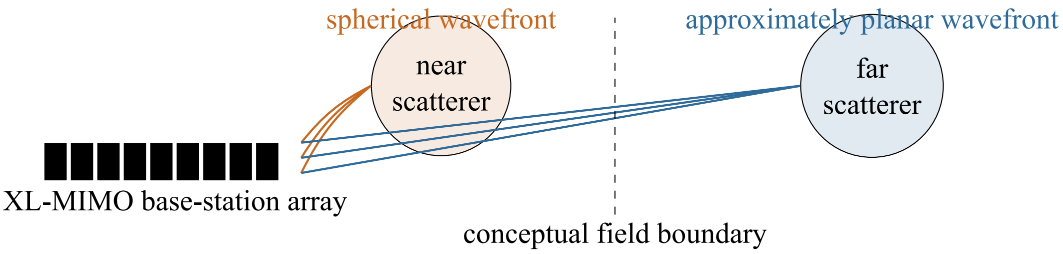

Figure 1: Original conceptual illustration of a hybrid-field XL-MIMO environment. Different paths can obey different wavefront models.

2 Problem and Contributions

Far-field methods exploit sparsity in an angle-domain dictionary, whereas near-field methods use a larger polar-domain dictionary indexed by angle and distance. Applying either one to the complete hybrid channel produces model mismatch and energy leakage. The paper addresses this gap through three linked contributions:

1. It identifies mixed far-field and near-field propagation as a realistic XL-MIMO channel feature.
2. It introduces a hybrid-field model whose mixing parameter controls the proportion of the two path types.
3. It proposes hybrid-field orthogonal matching pursuit (HF-OMP), which estimates each path type in its matching sparse domain while retaining low pilot overhead.

3 Signal Model and Propagation Regions

For downlink training from an N-antenna BS to a single-antenna user, the received pilots are

$$\boldsymbol{y} = \boldsymbol{P}\boldsymbol{h} + \boldsymbol{n}, \tag{1}$$

where $\boldsymbol{h}$ is the channel, $\boldsymbol{P}$ is the pilot matrix, and $\boldsymbol{n}$ is noise. Since low-overhead estimation requires $M \ll N$, recovery depends on a sparse physical model.

The boundary between propagation regions is approximated by the Rayleigh distance

$$Z = \frac{2D^2}{\lambda}, \tag{2}$$

with aperture D and wavelength λ. Scatterers beyond Z use a planar-wave model; those inside Z require a spherical-wave model. Because Z grows quadratically with D, XL-MIMO greatly enlarges the near field.

4 Existing Sparse Channel Models

4.1 Far-Field: Angle-Domain Sparsity

For L paths, planar-wave phases progress linearly across the array. The channel is represented as

$$\boldsymbol{h}_{\text{far}} = \boldsymbol{F}\boldsymbol{h}^{A}, \tag{3}$$

where $\boldsymbol{F}$ is a DFT dictionary and $\boldsymbol{h}^{A}$ is sparse because only a few angles contain paths. Angle-domain OMP exploits this sparsity.

4.2 Near-Field: Polar-Domain Sparsity

A spherical wave depends on angle and distance, so a DFT dictionary spreads its energy across many bins. The polar representation is

$$\boldsymbol{h}_{\text{near}} = \boldsymbol{W}\boldsymbol{h}^{P}, \tag{4}$$

where $\boldsymbol{W}$ samples angle–distance points and $\boldsymbol{h}^{P}$ is sparse. It is accurate for nearby scatterers, but $\boldsymbol{W}$ is larger and less orthogonal than $\boldsymbol{F}$.

Table 1: Comparison of the two conventional models.

	Far field	**Near field**	**Mismatch effect**
Wavefront	Planar	Spherical	Phase mismatch
Variables	Angle	Angle and distance	Energy spread
Sparse domain	DFT/angle	Polar	Extra atoms
Dictionary	$N \times N$	$N \times S$, $S > N$	Coherence and cost

For a mixed channel, near-field paths leak in $\boldsymbol{F}$ and far-field paths leak in $\boldsymbol{W}$. The complete channel is therefore not sparse enough in either single dictionary for accurate conventional OMP.

5 Proposed Hybrid-Field Channel Model

The paper models the channel as a superposition of far-field and near-field components:

$$\begin{aligned} \boldsymbol{h}_{\text{hybrid}} = & \sqrt{\frac{N}{L}} \sum_{\ell_f=1}^{\gamma L} \alpha_{\ell_f} \boldsymbol{a}(\theta_{\ell_f}) \\ & + \sqrt{\frac{N}{L}} \sum_{\ell_n=1}^{(1-\gamma)L} \beta_{\ell_n} \boldsymbol{b}(\theta_{\ell_n}, r_{\ell_n}), \end{aligned} \tag{5}$$

where L is the total number of paths and $\gamma \in [0, 1]$ controls their composition. The first sum contains γL planar-wave paths, and the second contains $(1-\gamma)L$ spherical-wave paths.

This formulation is general rather than a third unrelated propagation law. At $\gamma = 1$, it reduces to the standard far-field model. At $\gamma = 0$, it becomes the near-field model. Intermediate values describe mixed environments, such as a distant direct link accompanied by scatterers close to the BS.

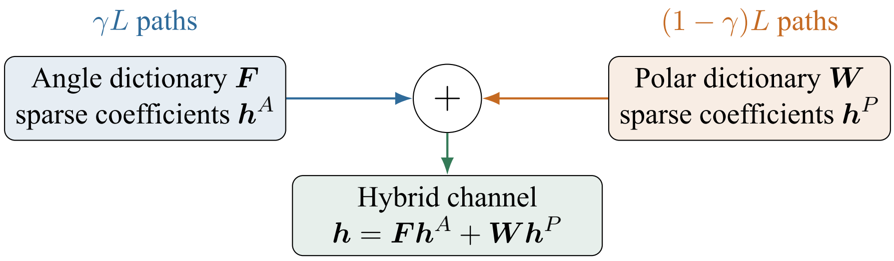

Figure 2: Original representation of the hybrid-field model as the sum of two components that are sparse in different dictionaries.

6 Why a Single Dictionary Fails

Substituting the two sparse representations into Equation (1) gives

$$\boldsymbol{y} = \boldsymbol{P}\boldsymbol{F}\boldsymbol{h}^A + \boldsymbol{P}\boldsymbol{W}\boldsymbol{h}^P + \boldsymbol{n}. \tag{6}$$

This equation exposes the estimation challenge. If far-field OMP uses only $\boldsymbol{P}\boldsymbol{F}$, the spherical-wave component behaves like structured interference and may require many angle atoms. If near-field OMP uses only $\boldsymbol{P}\boldsymbol{W}$, a planar-wave component can leak among multiple polar atoms. In either case, the assumed sparsity level and the true representation no longer agree.

7 Interpretation and Significance

The main conceptual contribution is to match each physical path class with its natural coordinate system. The hybrid model preserves the low number of physical scatterers instead of forcing every path into one oversized representation. It also provides a controlled simulation framework: varying γ tests pure near-field, mixed-field, and pure far-field conditions with the same total path count.

HF-OMP assumes that L and γ are known. A practical system would need to estimate or select both values adaptively.

8 HF-OMP Channel Estimation

HF-OMP separates Equation (6) into two sparse-recovery stages. It first searches the smaller, better-conditioned angle dictionary and then explains the remaining observation with the polar dictionary.

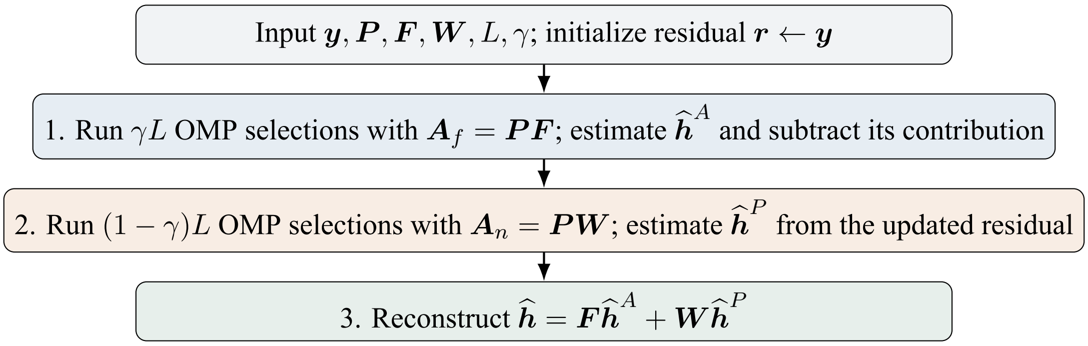

Figure 3: Original flowchart of the three-stage HF-OMP procedure.

8.1 Stage 1: Far-Field Recovery

With $\boldsymbol{A}_f = \boldsymbol{P}\boldsymbol{F}$, each OMP iteration selects the column most correlated with the current residual. A least-squares update estimates the gains on the selected support. After γL selections, the algorithm subtracts the reconstructed far-field contribution. Starting with this stage is computationally attractive because $\boldsymbol{F}$ has only N orthogonal columns.

8.2 Stage 2: Near-Field Recovery

The remaining residual is searched using $\boldsymbol{A}_n = \boldsymbol{P}\boldsymbol{W}$. The selected polar atoms identify both angle and distance. The residual update accounts for the estimated near-field component as well as the far-field component already found. At $\gamma = 1$ this stage disappears; at $\gamma = 0$ the first stage disappears, so conventional far-field and near-field OMP are endpoint cases of HF-OMP.

8.3 Stage 3: Reconstruction and Complexity

The two estimated sparse vectors are transformed back to the antenna domain and added. Following the paper's notation, the approximate complexity is

$$\mathcal{O}\big(NM(\gamma L)^3\big) + \mathcal{O}\big(SM((1-\gamma)L)^3\big) + \mathcal{O}(NS). \tag{7}$$

The polar search is the expensive part because S can greatly exceed N. HF-OMP nevertheless avoids using the large polar dictionary for paths that are naturally represented by the DFT dictionary.

9 Practical Trade-Offs

The sequential structure is simple and interpretable, but early support errors can contaminate the residual used in Stage 2. Grid-based dictionaries also create off-grid leakage, and closely spaced atoms in $\boldsymbol{W}$ can be difficult to distinguish. Joint recovery, adaptive field classification, or off-grid refinement could improve robustness at the price of additional computation.

10 Simulation Results

The simulations use the parameters in Table 2. With $M = 256 < N = 512$, the compressed-sensing schemes use half as many pilots as the MMSE benchmark ($M = 512$). HF-OMP is compared with far-field OMP, near-field OMP, and this full-dimensional benchmark.

Table 2: Principal simulation settings reported by Wei and Dai.

BS antennas	$N = 512$	Carrier frequency	30 GHz
Wavelength	$\lambda = 0.01$ m	Physical paths	$L = 6$
CS pilots	$M = 256$	Polar grid size	$S = 2071$
Path angle	uniform	Path distance	10–80 m

10.1 NMSE versus SNR

For $\gamma = 0.5$, HF-OMP achieves lower normalized mean-square error (NMSE) than either mismatched OMP baseline across the reported SNR range. Far-field OMP spreads spherical paths, while near-field OMP represents planar paths with a coherent, oversized dictionary. HF-OMP improves accuracy by recovering each component in its sparse domain. The MMSE curve is not a like-for-like low-overhead method because it uses twice as many pilots.

10.2 NMSE versus Field Composition

At 5 dB, the authors vary γ. HF-OMP matches near-field OMP at $\gamma = 0$ and far-field OMP at $\gamma = 1$, because those are its two endpoint cases. For mixed channels it performs better than either single-field estimator. Thus performance depends on modeling field composition, not only on the sparse-recovery algorithm.

11 Critical Assessment and Conclusion

The hybrid model clearly exposes mismatch loss, and HF-OMP adds an interpretable two-stage search to standard OMP. Its limitations are the assumptions of known L and γ, on-grid paths, and a limited simulation study. The large polar dictionary raises memory and computation, while far-first recovery can propagate errors.

The paper establishes that XL-MIMO should not be forced into one field model: domain-matched recovery improves NMSE at the same low pilot overhead. Future work should estimate γ, refine off-grid parameters, use joint or Bayesian recovery, and test wideband, multi-user, and measured channels.

References

[1] X. Wei and L. Dai, “Channel estimation for extremely large-scale massive MIMO: Far-field, near-field, or hybrid-field?” *IEEE Communications Letters*, vol. 26, no. 1, pp. 177–181, Jan. 2022, doi: 10.1109/LCOMM.2021.3124927.

[2] M. Cui and L. Dai, “Channel estimation for extremely large-scale MIMO: Far-field or near-field?” *IEEE Transactions on Communications*, vol. 70, no. 4, pp. 2663–2677, Apr. 2022.

Q&A

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